

DSA 210 Project Proposal

Analysis of Instagram Content Types and Engagement

Research Question

Does content type significantly affect engagement on Instagram, and is this effect influenced by user activity levels?

Motivation

Social media platforms, especially Instagram, play a central role in digital marketing and communication. As different content formats such as photos, reels, and carousels become more common, understanding which types generate higher engagement has become increasingly important. In addition, user activity patterns may influence how content performs. This project aims to analyze whether certain types of posts consistently outperform others and whether this effect depends on when the content is shared.

Hypothesis

The project hypothesizes that different Instagram content types (photo, reel, carousel) lead to significantly different engagement levels, and that posts shared during high user activity periods tend to receive higher engagement.

Data Sources

The primary dataset will be obtained from Kaggle and includes Instagram post-level data such as likes, comments, post type, and timestamps. The dataset contains approximately 29,000 observations, allowing for a detailed analysis of engagement patterns across different content types.

To enrich the analysis, a second dataset containing Instagram user behavior and activity patterns will be used. This dataset provides insights into general user activity trends, which will be used to identify peak and off-peak usage periods.

Data Preparation

The data will be cleaned and processed to extract relevant features such as engagement (likes and comments combined) and posting hour. Since the two datasets do not share common user identifiers, they will not be directly merged. Instead, the enrichment dataset will be used to derive general activity patterns. Based on these patterns, each post in the main dataset will be categorized as being shared during peak or off-peak hours.

Analysis Plan

The analysis will include exploratory data analysis (EDA), visualization of engagement patterns, and statistical testing to examine differences across content types and activity levels. The goal is to identify how content type and timing jointly influence engagement and to provide insights into more effective social media strategies.